
Is the Reconstruction–Forecasting Tradeoff a Frontier or an Operating Point?

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Abstract

1 Autoencoder-based temporal models are routinely trained on a weighted sum of a
2 reconstruction loss and a forecasting loss, with the weight treated as a hyperparam-
3 eter. This practice assumes the balance is a genuine frontier. We ask whether it is
4 instead a suboptimal operating point induced by forcing one latent to serve both ob-
5 jectives. On a controlled chaotic benchmark (Lorenz-63 lifted to 30 noisy channels,
6 known intrinsic dimension 3) we split the representation into an unbounded 3-d
7 carrier read only by the forecaster and a sigmoid-bounded 8×8 carrier read only
8 by the decoder, trained in two stages so that the forecast gradient provably cannot
9 reach the decoder. At matched parameters, matched wall clock and matched seeds,
10 the split Pareto-dominates *every* setting of the weighted-loss knob that forecasts
11 at all. We then report the result that complicates this: the advantage is compute-
12 dependent — given $10\times$ the wall clock, the balancing baseline overtakes it on both
13 headline axes. What does not transfer with compute is stability: bounded decoding
14 holds out-of-range rollouts at 0% against 100% for every unbounded baseline. The
15 tradeoff is therefore not one fixed frontier, and compute-matched comparison alone
16 cannot settle whether it is dissolvable.

17 1 Introduction

18 A large body of work trains one encoder to serve two objectives at once: reconstruct the present and
19 forecast the future. The standard mechanism is a weighted loss, $\mathcal{L} = (1 - \phi)\mathcal{L}_{\text{rec}} + \phi\mathcal{L}_{\text{fct}}$, with
20 ϕ tuned per dataset. Coupled Attention Networks [Xia et al., 2024] call their weights “a tradeoff
21 between the prediction and reconstruction models”; CAAE [Xie et al., 2026] names the “objective
22 mismatch between training for reconstruction fidelity and testing for discriminative decisions”;
23 reconstruction-plus-prediction hybrids fuse both objectives onto one shared latent, some with explicit
24 weights [Xiao et al., 2021], some by plainly summing the two losses [Cai et al., 2023]; and SSP [Lu
25 et al., 2026] pursues the same separation by spectral truncation, “decoupling reconstruction fidelity
26 from rollout regularity.” Across all of them the tradeoff is something to *tune*. Almost nobody asks
27 whether the tuning is necessary. The ingredients exist separately — bounded latents are routine in
28 representation learning [Mentzer et al., 2024], and Koopman-style forecasters constrain the latent
29 *dynamics* to buy long-horizon stability [Azencot et al., 2020, Liu et al., 2023] — but neither line
30 splits the two objectives across separate carriers, and the Koopman forecasters still run a single latent
31 through both.

32 **What we are not claiming.** Rate–distortion theory [Shannon, 1959, Berger, 1971] proves a frontier
33 between rate and distortion for a source, and the information bottleneck [Tishby et al., 2000] extends

this to predictive representations. We are not claiming to beat any such bound. Rate–distortion constrains distortion *at a fixed rate*; it says nothing about which point inside the achievable region a given architecture and schedule lands on. Our claim is only the latter: at fixed capacity and compute, a shared latent under a weighted loss does not reach the best available point, and separating the carriers moves it.

Contributions. (i) A controlled, compute-matched test of whether the tradeoff is a frontier or an operating point, on a system of known intrinsic dimension. (ii) A bounded/unbounded split Pareto-dominating every weighted-loss setting that forecasts at all, at matched parameters, wall clock and seeds. (iii) A negative result constraining this: the dominance does not survive a $10\times$ baseline budget. (iv) A bounded-shared-latent control isolating which ingredient buys what.

2 Method

Objects. For $x_t \in \mathbb{R}^n$ write $h_t = E(x_t) \in \mathbb{R}^n$,
 $b_t = f(h_t) \in \mathbb{R}^k$ and $C_t = m(b_t) \in [0, 1]^{a \times b}$ with a decoder $D : [0, 1]^{a \times b} \rightarrow \mathbb{R}^n$ a forecaster
 $g : \mathbb{R}^k \rightarrow \mathbb{R}^k$ and a teacher
 $r : \mathbb{R}^n \rightarrow [0, 1]^{a \times b}$ Here b is the *unbounded* carrier ($k = 3$) and C the *bounded* one ($a = b = 8$).
Reconstruction runs $x_t \rightarrow h_t \rightarrow b_t \rightarrow C_t \rightarrow \hat{x}_t$; forecasting steps only b and recomputes C from it:
 $b_{t+1} = g(b_t)$, $C_{t+1} = m(b_{t+1})$, $\hat{x}_{t+1} = D(C_{t+1})$.

Consistency. The chain is coherent only if the indirect route to the bounded latent agrees with the direct one, $m(f(h_t)) \approx r(h_t)$. As r sees all n dimensions and m only k , agreement forces f to retain everything C needs while compressing $30 \rightarrow 3$. r is a training-only teacher, never run at inference.

Two stages, and why the split is exact. Stage 1 trains $\theta_A = \{E, f, m, r, D\}$ on single timesteps:
 $\mathcal{L}_A = \|D(m(f(E(x_t)))) - x_t\|^2 + \lambda \|m(f(E(x_t))) - r(E(x_t))\|_F^2$. Stage 2 *freezes* θ_A , runs the frozen encoder over every training timestamp to obtain the series b_t , and trains g alone, free-running, entirely in b -space:

$$\mathcal{L}_B(\theta_g) = \frac{1}{H} \sum_{\tau=1}^H \|\hat{b}_{t+\tau} - b_{t+\tau}\|^2 \text{ with } \hat{b}_{t+\tau} = g^{\circ \tau}(b_t)$$

Neither m nor D appears in $\mathcal{L}_B$, so

$\partial \mathcal{L}_B / \partial \theta_A \equiv 0$: improving the forecast *cannot* degrade reconstruction. This is the decoupling, and it is an identity rather than a penalty. Contrast the weighted loss, where

$\partial \mathcal{L}_{\text{fcst}} / \partial \theta_{\text{enc}} \neq 0$ and ϕ merely chooses which objective loses.

Boundedness gives unconditional stability. m ends in a sigmoid, so $C_{t+\tau} \in [0, 1]^{a \times b}$ by construction. D is continuous on that compact set, hence $\|\hat{x}_{t+\tau}\| \leq \sup_C \|D(C)\| < \infty$ for all τ , *independently of* $\hat{b}_{t+\tau}$: however far the unbounded carrier drifts, the decoded trajectory cannot leave a bounded set, so error saturates instead of exploding. This costs zero parameters.

3 Experimental setup

Data. Lorenz-63 [Lorenz, 1963] at $dt = 0.01$, lifted through a frozen random tanh map to $n = 30$ channels with Gaussian noise at 5% of each channel’s spread. True intrinsic dimension is 3, so k is a probe, not a hyperparameter. Training pools 8 independent trajectories, each split 70/15/15 chronologically; 32 further trajectories are held out entirely. Every standardisation constant is fit on the pooled training portion only, and the reconstruction floor is *measured* from the realised noise draw (0.002492) rather than assumed as noise².

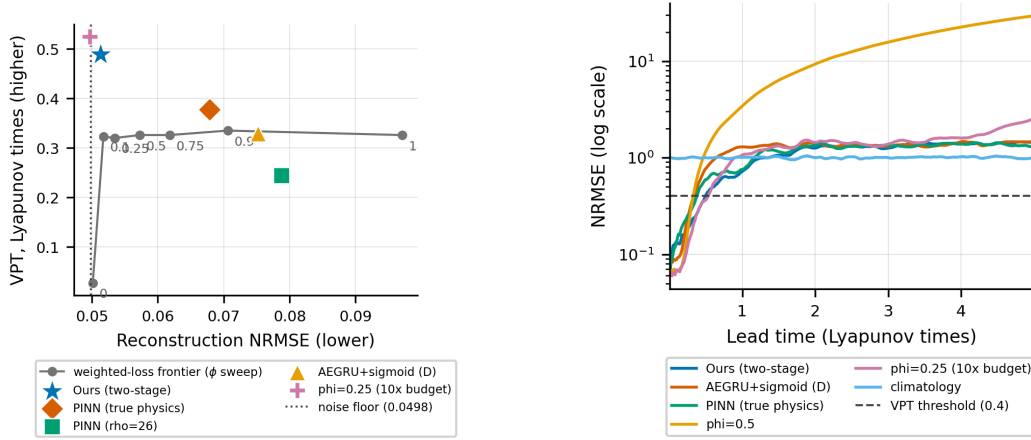


Figure 1: **Left:** the ϕ frontier (grey) is nearly flat — paying reconstruction buys almost no forecast — and the split (star) sits outside the region it reaches. The $10\times$ -budget point (pink) is the counterpoint: the frontier is not compute-invariant. **Right:** error vs lead time, log scale. Unbounded models climb without limit; every bounded decode path saturates at climatology (≈ 1).

Table 1: Matched parameters, wall clock and seeds where marked. $\times$ floor is reconstruction MSE over the measured floor (lower better); divergence is the fraction of rollouts leaving the training range. The $10\times$ row is an off-protocol control, not a matched comparison. Full table in Appendix A.

Model	seeds	$\times$ floor $\downarrow$	VPT $\uparrow$	NRMSE@5LT $\downarrow$	diverge $\downarrow$
Ours (two-stage)	3	1.06	0.489 $\pm$ 0.013	1.18	0%
$\phi=0$ (no forecasting)	1	1.01	0.027	30.6	100%
$\phi=0.1$	3	1.08	0.323 $\pm$ 0.011	15.5	100%
$\phi=0.5$	3	1.32	0.326 $\pm$ 0.013	15.8	100%
AEGRU+sigmoid (D)	3	2.28	0.329 $\pm$ 0.004	1.28	1.6%
PINN (true physics)	3	1.87	0.378 $\pm$ 0.019	1.19	26%
persistence / climatology	—	—	0.036 / 0.000	1.37 / 1.00	0%
$\phi=0.25$, 10 $\times$ budget	1	0.99	0.526	1.43	22%

Baselines. (B) weighted-loss AE+GRU, $\phi \in \{0, 0.1, 0.25, 0.5, 0.75, 0.9, 1\}$, 16-d shared latent; (D) the same with its *shared* latent sigmoid-bounded through rollout — the control holding boundedness fixed while removing the split; (C) a physics-informed AE whose 3-d latent is calibrated to true Lorenz coordinates and rolled out by RK4 on the known equations, plus a misspecified $\rho = 26$ variant; and persistence/climatology.

Protocol. Every model gets **124,482 trainable parameters and 120 s of wall clock on one pinned CPU thread**. Ours, the PINN, $\phi \in \{0.1, 0.25, 0.5\}$ and model D run 3 seeds; remaining ϕ settings run 1. We report reconstruction MSE over the measured floor; VPT (lead time at which normalised error first crosses 0.4, in Lyapunov times [Vlachas et al., 2020]); error at 5 Lyapunov times; and *divergence*, the fraction of rollouts leaving the training box.

4 Results

At matched compute, the split dominates. The split reaches $1.06\times$ the floor at VPT 0.489. Every ϕ that forecasts at all is worse on *both* axes at once: the best reaches 0.335 ($\phi=0.9$, App. A), and the only one reconstructing better ($\phi=0$, $1.01\times$) has VPT 0.027, below persistence — it does not forecast. The split is thus not another point on the knob’s curve but outside the region it reaches.

89 That curve is itself nearly flat (0.320–0.335 across $\phi \in [0.1, 1]$) while reconstruction degrades $3.8\times$:
90 paying reconstruction buys almost no forecast here.

91 **The negative result: this is compute-dependent.** Held at 120 s the above reads as dissolution; it is
92 not that simple. Given $10\times$ the wall clock, $\phi=0.25$ reaches VPT 0.526 at $0.99\times$ the floor — better
93 than the split on both headline axes. The shift ($\Delta\text{VPT} = 0.205$) is an order of magnitude larger than
94 the $1\times$ run’s 3-seed spread ($\sigma = 0.015$), so the family is *measurably undertrained* at that budget, not
95 frontier-limited there. This bounds the claim: a compute-matched Pareto comparison, the standard
96 protocol here, can invert under a compute rescaling. Any claim that architecture X dissolves the
97 tradeoff, ours included, is only as strong as the budget it was measured at.

98 **What does not follow compute: stability.** Divergence separates the models in a way compute does
99 not fix. Every unbounded baseline leaves the training range on 100% of rollouts at $1\times$; at $10\times$ that
100 falls only to 22%. The split is at 0%, with error at 5 Lyapunov times of 1.18 — the climatology level,
101 i.e. saturating rather than exploding, as the boundedness argument predicts. That is a guarantee from
102 the function class, not something better optimisation supplies.

103 **Which ingredient buys what.** Model D holds boundedness fixed and removes the split. It reaches
104 divergence 1.6% — essentially the split’s stability — but reconstruction $2.28\times$ floor and VPT 0.329,
105 no better than the knob. The two ingredients buy different things: **boundedness buys stability**;
106 **separating the carriers buys the reconstruction and forecast quality.** Neither alone reproduces the
107 result — our cleanest evidence that the split does real work rather than acting as a reparameterised ϕ .

108 **Schedule, noise, physics.** End-to-end training of the same modules degrades reconstruction $4.4\times$
109 and VPT to 0.372: the schedule, not just the architecture, is load-bearing. At noise 0.15/0.30 (1
110 seed) the split still leads the best ϕ — VPT gaps $+0.163, +0.290, +0.208$ — but *not monotonically*,
111 peaking at 0.15; we report the sequence, not a direction. The PINN needed a repair to be a fair
112 comparator: its decoder reads the uncalibrated latent, so a *relative* physics residual was satisfiable
113 in physically wrong coordinates. Recalibrating each epoch moved its forecast loss $17.6 \rightarrow 0.015$
114 and its state-recovery R^2 from $[0.64, 0.70, 0.10]$ to $[0.996, 0.989, 0.981]$; post-fix it has the best
115 50-Lyapunov-time attractor fidelity here (Wasserstein 0.068 vs the split’s 0.170). Appendices B–D.

116 5 Limitations

117 **One system:** Lorenz-63, one lift; the cross-domain phenomenon claim motivating this work is
118 untested here. **Compute-dependence unresolved:** we did not run $10\times$ with 3 seeds, locate the
119 crossover, or give the split $10\times$. **Fixed capacity:** the bounded carrier is a fixed 8×8 grid and we
120 never varied it, so how much of the result rests on that particular size is unknown. **Uneven seeds:**
121 single-seed rows report $\sigma = 0$ because $n = 1$. **A floor artifact:** $\phi=0.25$ at $10\times$ sits at $0.99\times$ the
122 measured floor; a capacity-matched control (latent=3) clears it, indicating a wide unregularised
123 latent exploiting cross-channel redundancy rather than a leak. **Constructed baselines:** the ϕ sweep
124 and model D are the balancing recipe in its plainest form, not named published methods.

125 6 Conclusion

126 At matched parameters, wall clock and seeds, separating a bounded decoder-facing carrier from an
127 unbounded forecaster-facing one reaches an operating point no setting of the knob reaches, on a
128 curve that is itself nearly flat. But that advantage is compute-dependent: we show it inverting at $10\times$
129 budget. What does not depend on compute is stability — bounded decoding gives 0% divergence by
130 construction where every unbounded baseline diverges on every rollout. The honest conclusion is
131 narrower than “the tradeoff dissolves”: it is not a single fixed frontier, compute-matched comparison
132 alone cannot settle whether it is dissolvable, and the reliability half of the problem is separable from
133 the accuracy half.

References

- O. Azencot, N. B. Erichson, V. Lin, and M. W. Mahoney. Forecasting sequential data using consistent Koopman autoencoders. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, pages 475–485, 2020.
- T. Berger. *Rate Distortion Theory: A Mathematical Basis for Data Compression*. Prentice-Hall, Englewood Cliffs, NJ, 1971.
- B. Cai, S. Yang, L. Gao, and Y. Xiang. Hybrid variational autoencoder for time series forecasting. *Knowledge-Based Systems*, 281:111079, 2023. arXiv:2303.07048.
- Y. Liu, C. Li, J. Wang, and M. Long. Koopa: Learning non-stationary time series dynamics with Koopman predictors. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2305.18803.
- E. N. Lorenz. Deterministic nonperiodic flow. *Journal of the Atmospheric Sciences*, 20(2):130–141, 1963.
- X. Lu, Y. Yuan, and J. Shi. Stable long-horizon PDE forecasting via latent structured spectral propagators. *arXiv preprint arXiv:2605.10154*, 2026.
- F. Mentzer, D. Minnen, E. Agustsson, and M. Tschannen. Finite scalar quantization: VQ-VAE made simple. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2309.15505.
- C. E. Shannon. Coding theorems for a discrete source with a fidelity criterion. *IRE National Convention Record*, part 4, pages 142–163, 1959.
- N. Tishby, F. C. Pereira, and W. Bialek. The information bottleneck method. *arXiv preprint physics/0004057*, 2000.
- P. R. Vlachas, J. Pathak, B. R. Hunt, T. P. Sapsis, M. Girvan, E. Ott, and P. Koumoutsakos. Back-propagation algorithms and reservoir computing in recurrent neural networks for the forecasting of complex spatiotemporal dynamics. *Neural Networks*, 126:191–217, 2020.
- F. Xia, X. Chen, S. Yu, M. Hou, M. Liu, and L. You. Coupled attention networks for multivariate time series anomaly detection. *IEEE Transactions on Emerging Topics in Computing*, 12:240–253, 2024. arXiv:2306.07114.
- Q. Xiao, S. Shao, and J. Wang. Memory-augmented adversarial autoencoders for multivariate time-series anomaly detection with deep reconstruction and prediction. *arXiv preprint arXiv:2110.08306*, 2021.
- X. Xie, K. Liu, Y. Wang, M. Wu, H. Zhang, and T. Wan. CAAE: Contrastive adversarial autoencoder for multivariate time series anomaly detection. *Pattern Recognition*, 2026.

166 A Full results

Table 2: Complete comparison, all 14 configurations. Epochs are seed-0 counts within the 120 s budget (Ours reports stage 1 + stage 2 combined).

Model	seeds	epochs	recon MSE	NRMSE	VPT ($\pm\sigma$)	diverge
$\phi=0.25$ (10 $\times$)	1	19	0.0025	0.0497	0.526	0.219
Ours (two-stage)	3	16	0.0026	0.0513	0.489 ± 0.013	0.000
PINN (true physics)	3	16	0.0047	0.0678	0.378 ± 0.019	0.260
$\phi=0.9$	1	2	0.0050	0.0706	0.335	1.000
AEGRU+sigmoid (D)	3	2	0.0057	0.0752	0.329 ± 0.004	0.016
$\phi=0.75$	1	2	0.0038	0.0618	0.326	1.000
$\phi=1$	1	2	0.0095	0.0971	0.326	1.000
$\phi=0.5$	3	2	0.0033	0.0572	0.326 ± 0.013	1.000
$\phi=0.1$	3	2	0.0027	0.0517	0.323 ± 0.011	1.000
$\phi=0.25$	3	2	0.0029	0.0535	0.320 ± 0.015	1.000
PINN ($\rho=26$)	1	16	0.0062	0.0788	0.245	0.000
persistence	—	—	—	—	0.036	0.000
$\phi=0$	1	2	0.0025	0.0501	0.027	1.000
climatology	—	—	—	—	0.000	0.000

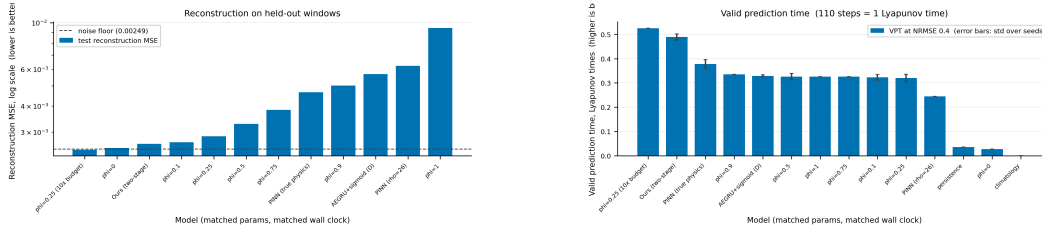


Figure 2: The two headline axes as bar charts. Left: reconstruction against the measured noise floor (dashed). Right: valid prediction time, error bars are the standard deviation over seeds — bars without one are $n = 1$.

167 Note the epochs column: the ϕ models complete ~ 2 epochs against Ours' 16 in the same wall clock,
168 because the joint model re-encodes full sequences every step while stage 1 trains on single timesteps
169 and stage 2 on precomputed latents. Equal seconds is not equal gradient steps; this is the mechanism
170 behind the budget result of §4, and it is a property of the architectures, not a protocol error.

171 B Long-horizon attractor statistics (50 Lyapunov times)

Table 3: Attractor statistics after 5,500 free-running steps. Wasserstein is the mean 1-D distance between per-channel marginals of the rollout and of the truth (lower better).

Model	Wasserstein ↓	spectrum log-err ↓	diverge	final NRMSE
PINN (true physics)	0.068	1.88	0.906	1.42
PINN ($\rho=26$)	0.149	2.07	0.188	1.42
Ours (two-stage)	0.170	2.06	0.000	1.44
AEGRU+sigmoid (D)	0.568	3.14	0.000	1.46
$\phi=1$	139.3	8.24	1.000	389.7
$\phi=0.5$	170.1	8.32	1.000	417.9
$\phi=0$	213.1	8.59	1.000	534.3

172 After 5,500 blind steps the split still traces the attractor (Wasserstein 0.170) while every weighted-loss
 173 setting is off by three orders of magnitude. The repaired PINN is best on this axis, as expected of a
 174 model handed the exact equations.

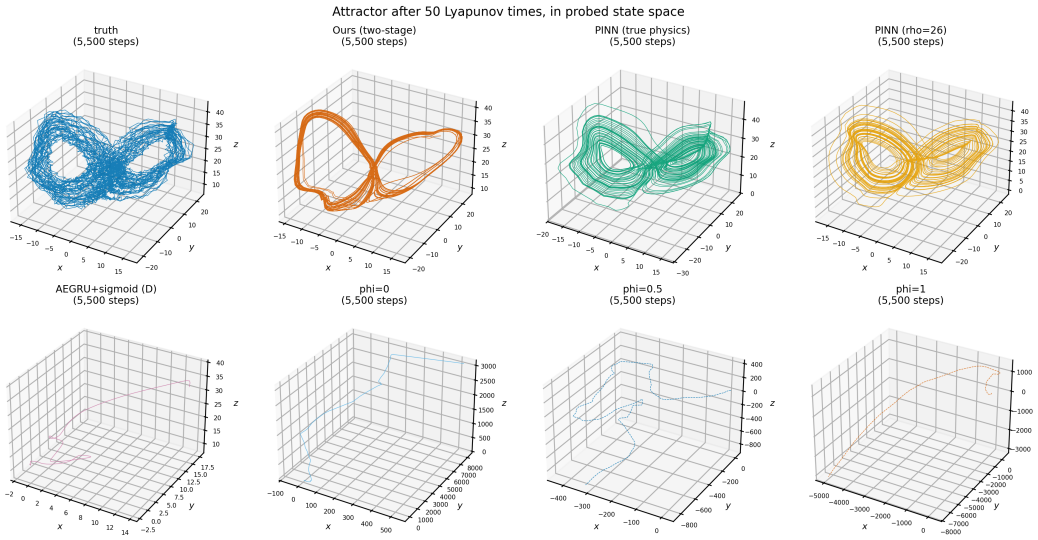


Figure 3: Attractor after 50 Lyapunov times (5,500 free-running steps), in probed state space. Truth first. The split and both PINNs still trace a recognisable butterfly; the weighted-loss settings have left the attractor entirely.

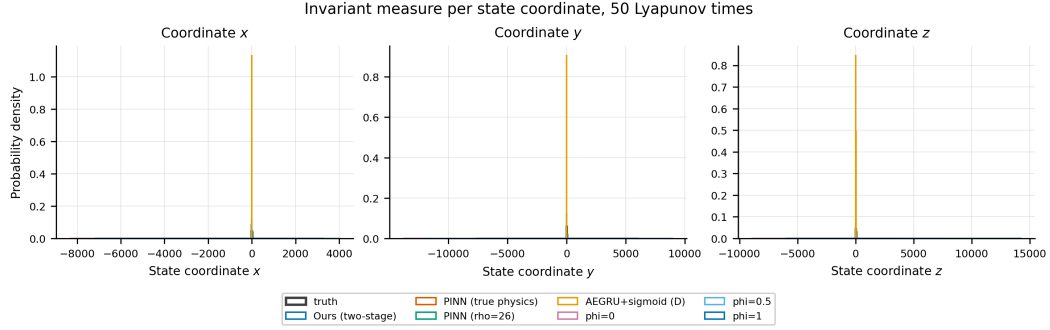


Figure 4: Invariant measure per state coordinate at 50 Lyapunov times — the distribution the Wasserstein column scores. Curves that collapse to a spike, or sit far off truth, correspond to rollouts that have diverged.

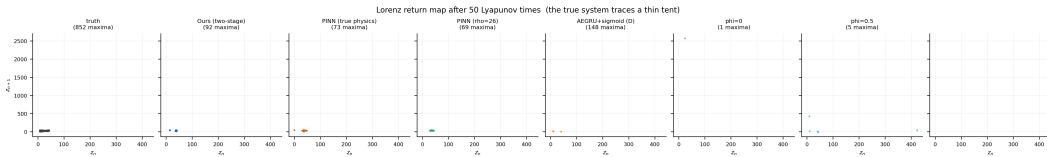


Figure 5: Lorenz return map (successive maxima of z) after 50 Lyapunov times. The true system traces a thin tent; a model with the right climate but the wrong geometry produces a visibly fattened or collapsed one.

175 C Noise sweep

Table 4: Headline comparison re-run at three observation-noise levels; one seed at 0.15 and 0.30.

noise	model	$\times$ floor	VPT	diverge
0.05	Ours	1.06	0.489	0.000
0.05	$\phi=0.1 / \phi=0.5$	1.08 / 1.32	0.323 / 0.326	1.000
0.05	AEGRU+sigmoid (D)	2.28	0.329	0.016
0.15	Ours	0.93	0.598	0.000
0.15	$\phi=0.1 / \phi=0.5$	0.86 / 0.95	0.299 / 0.308	1.000
0.15	AEGRU+sigmoid (D)	1.08	0.290	0.000
0.30	Ours	0.92	0.462	0.000
0.30	$\phi=0.1 / \phi=0.5$	0.55 / 0.86	0.254 / 0.254	0.000 / 1.000
0.30	AEGRU+sigmoid (D)	0.92	0.245	0.000

176 At noise 0.15 and 0.30 several models fall below $1.0\times$ the measured floor. This is the same capacity
 177 artifact discussed in the limitations, amplified: with more noise there is more to denoise, and cross-
 178 channel redundancy lets a wide latent recover the clean signal slightly better than the per-channel
 179 floor implies. It affects all models equally and does not change the ordering.

180 D Boundedness audit and training cost

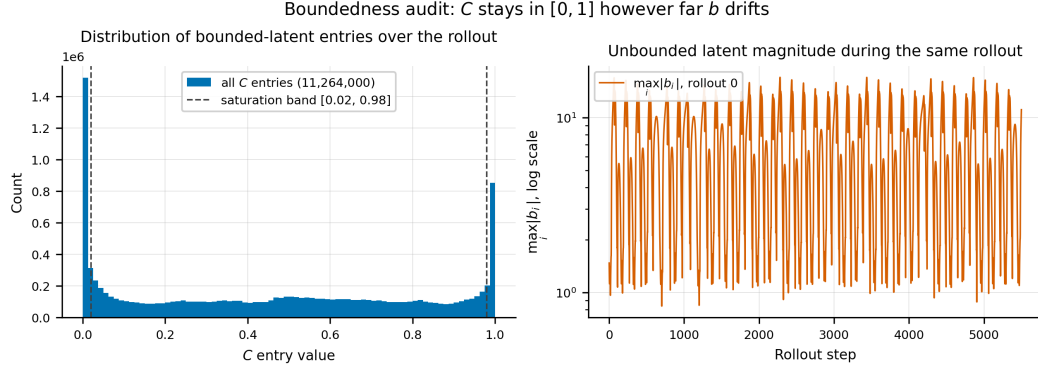


Figure 6: Left: distribution of bounded-latent entries over a 50-Lyapunov-time rollout — every value inside $[0, 1]$, as the construction guarantees. Right: the unbounded carrier’s magnitude over the same rollout. The bound holds on C regardless of what b does, which is the mechanism behind the 0% divergence in Table 1.

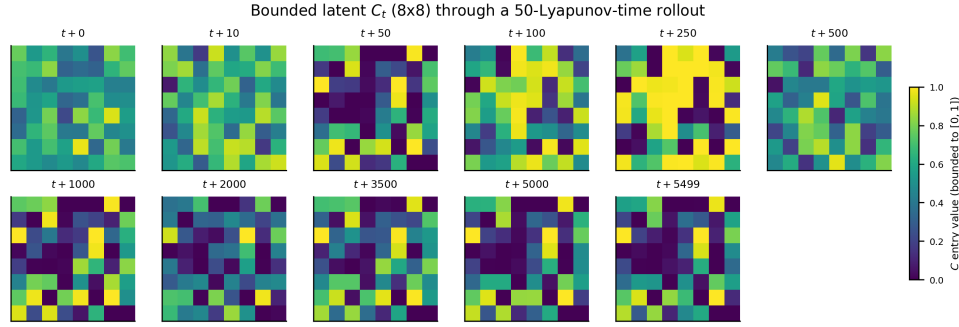


Figure 7: The bounded latent C_t (8×8) sampled through a 50-Lyapunov-time rollout. It neither saturates to the rails nor escapes the box.

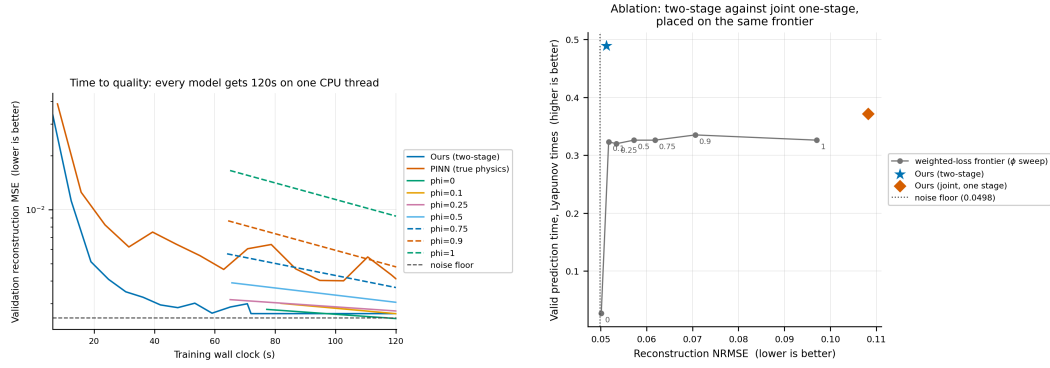


Figure 8: Left: validation reconstruction against wall clock — the split reaches $2\times$ the noise floor in 24.8 s against 63–120 s for the baselines, because stage 2 trains on precomputed latents with no decoder in the graph. Right: the two-stage schedule against end-to-end training of the identical modules, placed on the same frontier.

181 **E Reproducibility**

182 Code, data generator and the executed notebook (all outputs stored) accompany this submission. All
183 numbers come from a single end-to-end run at `QUICK=False` on one pinned CPU thread; the full
184 run takes approximately three hours from an empty checkpoint directory. Every model is cached
185 after training, so re-running reproduces the tables without retraining. Architecture: E is an MLP
186 $30 \rightarrow 128 \rightarrow 128 \rightarrow 30$; f is $30 \rightarrow 128 \rightarrow 64 \rightarrow 3$; m is $3 \rightarrow 64 \rightarrow 128 \rightarrow 64$ with a sigmoid; r is
187 $30 \rightarrow 128 \rightarrow 128 \rightarrow 64$ with a sigmoid; D is $64 \rightarrow 128 \rightarrow 128 \rightarrow 30$; g is a residual GRU ($3 \rightarrow 64$) with
188 a linear head. All activations GELU. Adam, $\eta = 10^{-3}$, $\lambda = 1$, stage-1 batch 256, stage-2 batch 64,
189 warm-up 64 steps, training rollout 20 steps, evaluation horizon 550 steps (5 Lyapunov times).